UNIT 6

RAY OPTICS

There are two ways of spreading light: to be the candle or the mirror that reflects it.
— Edith Wharton

Learning Objectives

In this unit, the students are exposed to,

- The ray aspect of light.
- The phenomenon of reflection and refraction of light.
- The determination of speed of light.
- The concepts related to mirrors, lenses, prisms etc.
- The different phenomena like dispersion and scattering of light.

6.1 INTRODUCTION

Light is mystical. Yet, its behaviour is so fascinating. It is difficult to comprehend light as a single entity. The *ray optics* deals with light that is represented as a ray travelling in straight lines. Here, the geometrical constructs get the permanence to understand some of the characteristics of light and the phenomena associated with it. There are several other phenomena which can only be explained using *wave optics*, which we study in the next Unit. There is also a quantum aspect of light which we can study as *quantum optics* in graduate level courses.

6.1.1 Ray optics

Light travels in a straight line in a medium. Light may deviate in its path only when it encounters the boundary of another medium (or) an obstacle. A ray of light gives information only about the direction of light. It does not give information about the other characteristics like intensity and colour of light. However, a ray is a sensible representation of light in ray optics. The path of the light is called a ray of light and a bundle of such rays is called a beam of light. In this chapter, we can explain the phenomena of reflection, refraction, dispersion and scattering of light, using the ray depiction of light.

6.1.2 Reflection

The bouncing back of light into the same medium when it encounters a reflecting surface is called *reflection* of light. Any polished surface can reflect light. Mirrors which are silver coated at their back can reflect almost 90% of the light falling on them. The angle of incidence *i* and the angle of reflection *r* are measured with respect to the normal drawn to the surface at the point

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